

PuTTY/HDFS Workflow for Iris Classification

Supporting Document

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1. Environment Setup in PuTTY

- Login to PuTTY with your sandbox credentials (maria_dev).
- Create a folder for the Iris dataset in HDFS:

```
hdfs dfs -mkdir /user/maria_dev/IRIS
```

- Verify the folder:

```
hdfs dfs -ls /user/maria_dev
```

2. Upload Dataset

- Download Iris dataset into sandbox local directory:

```
wget https://raw.githubusercontent.com/uiuc-cse/data-fa14/gh-pages/data/iris.csv -O  
/home/maria\_dev/IRIS/iris.csv
```

- Confirm file exists locally:

```
ls /home/maria_dev/IRIS
```

- Move dataset into HDFS:

```
hdfs dfs -copyFromLocal /home/maria_dev/IRIS/iris.csv /user/maria_dev/IRIS/iris.csv
```

- Verify upload:

```
hdfs dfs -ls /user/maria_dev/IRIS
```

3. Create Spark Script (iris_model.py)

- Open file in PuTTY:

```
vi iris_model.py
```

- Paste tuned classification script (Decision Tree, Random Forest, Logistic Regression with CrossValidator).

#####code:

```
from pyspark.sql import SparkSession
```

```
from pyspark.ml.feature import StringIndexer, VectorAssembler
```

```
from pyspark.ml.classification import DecisionTreeClassifier, RandomForestClassifier,  
LogisticRegression
```

```
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
```

```
from pyspark.ml.tuning import CrossValidator, ParamGridBuilder
```

```
# Step 1: Start Spark session
```

```
spark = SparkSession.builder.appName("IrisTunedComparison").getOrCreate()
```

```
# Step 2: Load dataset from HDFS
```

```
data = spark.read.csv("hdfs:///user/maria_dev/IRIS/iris.csv", header=True,  
inferSchema=True)
```

```
# Step 3: Preprocess
```

```
indexer = StringIndexer(inputCol="species", outputCol="label")
```

```
data = indexer.fit(data).transform(data)
```

```
assembler = VectorAssembler(  
    inputCols=["sepal_length", "sepal_width", "petal_length", "petal_width"],  
    outputCol="features"
```

```
)
```

```
data = assembler.transform(data)
```

```
# Step 4: Fixed train/test split
```

```
train, test = data.randomSplit([0.8, 0.2], seed=1)
```

```
# Step 5: Evaluators
```

```
evaluator_acc = MulticlassClassificationEvaluator(labelCol="label",  
predictionCol="prediction", metricName="accuracy")
```

```
evaluator_f1 = MulticlassClassificationEvaluator(labelCol="label",  
predictionCol="prediction", metricName="f1")
```

```
def tune_and_eval(model, paramGrid, train, test, name):
```

```
    cv = CrossValidator(estimator=model,
```

```
                        estimatorParamMaps=paramGrid,
```

```
                        evaluator=evaluator_acc,
```

```
                        numFolds=5)
```

```
    cvModel = cv.fit(train)
```

```
    preds = cvModel.bestModel.transform(test)
```

```
    acc = evaluator_acc.evaluate(preds)
```

```
    f1 = evaluator_f1.evaluate(preds)
```

```
    print("{} Accuracy: {:.2f}".format(name, acc))
```

```
    print("{} F1-score: {:.2f}".format(name, f1))
```

```
    return acc, f1
```

```
results = []
```

```
# Logistic Regression tuning
```

```
lr = LogisticRegression(featuresCol="features", labelCol="label")
```

```
lr_paramGrid = (ParamGridBuilder()
```

```
.addGrid(lr.regParam, [0.01, 0.1, 0.5])  
.addGrid(lr.elasticNetParam, [0.0, 0.5, 1.0])  
.addGrid(lr.maxIter, [50, 100])  
.build()  
lr_acc, lr_f1 = tune_and_eval(lr, lr_paramGrid, train, test, "Tuned Logistic Regression")  
results.append(("Logistic Regression", lr_acc, lr_f1))
```

Decision Tree tuning

```
dt = DecisionTreeClassifier(featuresCol="features", labelCol="label")  
dt_paramGrid = (ParamGridBuilder()  
    .addGrid(dt.maxDepth, [2, 4, 6, 8])  
    .addGrid(dt.maxBins, [16, 32, 64])  
    .build())  
dt_acc, dt_f1 = tune_and_eval(dt, dt_paramGrid, train, test, "Tuned Decision Tree")  
results.append(("Decision Tree", dt_acc, dt_f1))
```

Random Forest tuning

```
rf = RandomForestClassifier(featuresCol="features", labelCol="label")  
rf_paramGrid = (ParamGridBuilder()  
    .addGrid(rf.numTrees, [10, 20, 50])  
    .addGrid(rf.maxDepth, [4, 6, 8])  
    .build())  
rf_acc, rf_f1 = tune_and_eval(rf, rf_paramGrid, train, test, "Tuned Random Forest")  
results.append(("Random Forest", rf_acc, rf_f1))
```

Step 6: Print comparison table

```

print("\n=== Tuned Model Comparison ===")

print("{:<20} {:<15} {:<15}".format("Model", "Accuracy", "F1-score"))

for name, acc, f1 in results:

    print("{:<20} {:<15.2f} {:<15.2f}".format(name, acc, f1))


# Stop Spark

spark.stop()

#####end_code

```

4. Run Spark Script

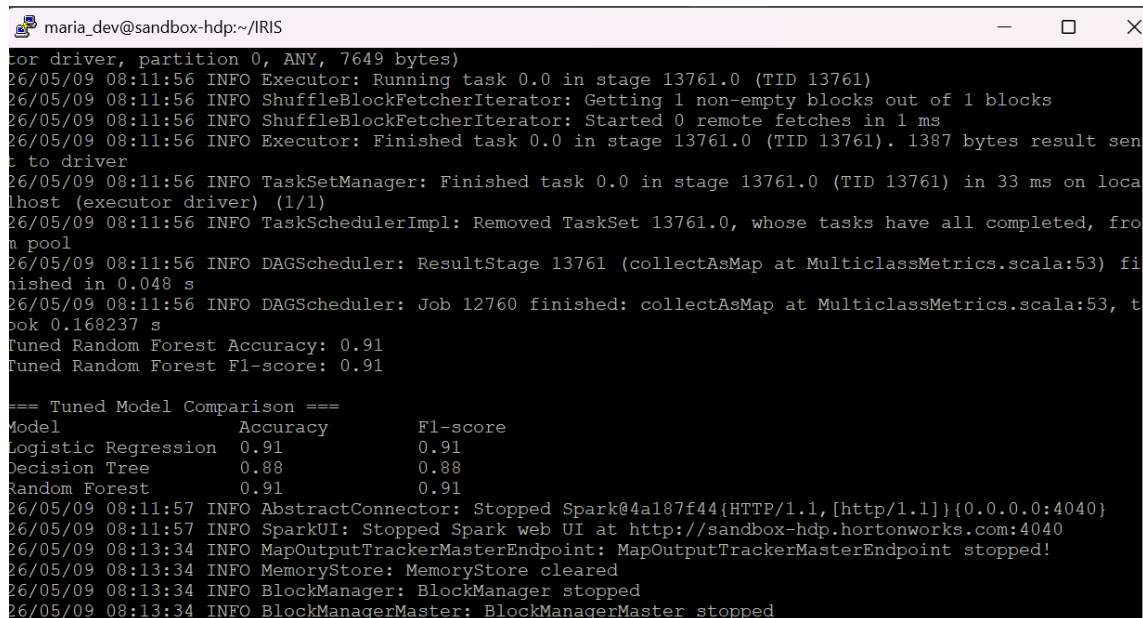
- Navigate to folder:

```
cd /home/maria_dev/IRIS/
```

- Execute script:

```
spark-submit iris_model.py
```

You should get these results:



```

maria_dev@sandbox-hdp:~/IRIS
For driver, partition 0, ANY, 7649 bytes)
26/05/09 08:11:56 INFO Executor: Running task 0.0 in stage 13761.0 (TID 13761)
26/05/09 08:11:56 INFO ShuffleBlockFetcherIterator: Getting 1 non-empty blocks out of 1 blocks
26/05/09 08:11:56 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 1 ms
26/05/09 08:11:56 INFO Executor: Finished task 0.0 in stage 13761.0 (TID 13761). 1387 bytes result sent to driver
26/05/09 08:11:56 INFO TaskSetManager: Finished task 0.0 in stage 13761.0 (TID 13761) in 33 ms on localhost (executor driver) (1/1)
26/05/09 08:11:56 INFO TaskSchedulerImpl: Removed TaskSet 13761.0, whose tasks have all completed, from pool
26/05/09 08:11:56 INFO DAGScheduler: ResultStage 13761 (collectAsMap at MulticlassMetrics.scala:53) finished in 0.048 s
26/05/09 08:11:56 INFO DAGScheduler: Job 12760 finished: collectAsMap at MulticlassMetrics.scala:53, took 0.168237 s
Tuned Random Forest Accuracy: 0.91
Tuned Random Forest F1-score: 0.91

=== Tuned Model Comparison ===
Model                Accuracy    F1-score
Logistic Regression   0.91        0.91
Decision Tree         0.88        0.88
Random Forest         0.91        0.91
26/05/09 08:11:57 INFO AbstractConnector: Stopped Spark@4a187f44[HTTP/1.1,[http/1.1]]{0.0.0.0:4040}
26/05/09 08:11:57 INFO SparkUI: Stopped Spark web UI at http://sandbox-hdp.hortonworks.com:4040
26/05/09 08:13:34 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/05/09 08:13:34 INFO MemoryStore: MemoryStore cleared
26/05/09 08:13:34 INFO BlockManager: BlockManager stopped
26/05/09 08:13:34 INFO BlockManagerMaster: BlockManagerMaster stopped

```